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Image forming apparatus

Abstract

An image forming apparatus to form an image on a recording material includes an apparatus main body having a first end provided with an opening and having a second end, a fixing device to heat the recording material, an opening and closing member, a drawer unit having a first photosensitive drum, and a transfer device. When being moved from a first inner position inside the apparatus main body to a first outer position outside the apparatus main body through the main body opening, the drawer unit is to be moved in a moving direction intersecting with a rotation axis of the first photosensitive drum, and is to be moved away from the second end. When being moved from a second inner position inside the apparatus main body to a second outer position outside the apparatus main body, the transfer device is moved away from the second end.

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Background/Summary

BACKGROUND

Field

(1) The present disclosure relates to an image forming apparatus that forms an image on a recording material.

Description of the Related Art

(2) Some image forming apparatuses, including a printer, a copying machine, and a multifunction peripheral, may include cartridges that are detachably attached to a tray that moves between inside and outside the apparatus main body.

(3) Japanese Patent Application Laid-Open No. 2010-244018 discusses an image forming apparatus including a tray to which cartridges are detachably attached and a fixing device. The tray can be moved out of the apparatus main body of the image forming apparatus through a plurality of openings.

(4) Japanese Patent Application Laid-Open No. 2015-206897 discusses an image forming apparatus including a tray to which cartridges are detachably attached, a transfer unit, and a fixing device. According to Japanese Patent Application Laid-Open No. 2015-206897, the tray and the transfer unit can be moved from inside to outside the image forming apparatus. Specifically, the fixing device is disposed at one end side of the apparatus main body of the image forming apparatus. When the transfer unit is moved out of the apparatus main body, the transfer unit moves in a direction from the other end side to the one end side of the apparatus main body. By contrast, when the tray is moved out of the apparatus main body, the tray is moved in a direction from the one end side to the other end side of the apparatus main body.

SUMMARY

(5) The present disclosure is directed to further developing the conventional technique.

(6) According to an aspect of the present disclosure, an image forming apparatus configured to form an image on a recording material includes an apparatus main body including a first end provided with a main body opening and a second end opposite to the first end in a horizontal direction, a fixing device configured to heat the recording material, an opening and closing member attached to the apparatus main body, wherein the opening and closing member is configured to be moved from a closed position, where the main body opening is covered, to an open position where the main body opening is exposed so that the opening and closing member separates from the fixing device, a drawer unit including (i) a first cartridge, (ii) a support member to which the first cartridge is detachably attached, and (iii) a first photosensitive drum disposed on the support member or the first cartridge, wherein the drawer unit is configured to be moved to a first inner position inside the apparatus main body and to a first outer position outside the apparatus main body, and a transfer device configured to be moved to a second inner position inside the apparatus main body and to a second outer position outside the apparatus main body and including a belt wherein the belt is configured to contact the first photosensitive drum with the transfer device at the second inner position, wherein the fixing device is closer to the first end than to the second end, wherein, when being moved from the first inner position to the first outer position through the main body opening, the drawer unit is configured to be moved in a moving direction intersecting with a rotation axis of the first photosensitive drum, and be moved away from the second end, and wherein, when being moved from the second inner position to the second outer position, the transfer device is configured to be moved away from the second end.

(7) Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates the overall configuration of a printer according to a first exemplary embodiment.
- (2) FIG. 2 is a diagram illustrating a printer according to the first exemplary embodiment with a door open.
- (3) FIG. 3 is a diagram illustrating the printer according to the first exemplary embodiment with a fixing device moved.
- (4) FIG. 4 is a diagram illustrating the printer according to the first exemplary embodiment when a transfer unit and a tray unit are pulled out.
- (5) FIG. 5 is a diagram illustrating the printer according to the first exemplary embodiment with the transfer unit alone pulled out.
- (6) FIG. 6 illustrates a movement of the transfer unit and the tray unit according to the first exemplary embodiment.
- (7) FIG. 7 illustrates the movement of the transfer unit and the tray unit according to the first exemplary embodiment.
- (8) FIG. 8 illustrates a coupling member that couples a tray of the tray unit with the transfer unit according to the first exemplary embodiment.
- (9) FIG. 9 illustrates the coupling member that couples the tray of the tray unit with the transfer unit according to the first exemplary embodiment.
- (10) FIG. 10 illustrates the coupling member that couples the tray of the tray unit with the transfer unit according to the first exemplary embodiment.
- (11) FIGS. 11A to 11D illustrate a tray unit according to modifications.
- (12) FIG. 12 illustrates the overall configuration of a printer according to a second exemplary embodiment.
- (13) FIG. 13 is a diagram illustrating the printer according to the second exemplary embodiment with a door, a fixing device, and a transfer unit moved.
- (14) FIG. 14 is a diagram illustrating the printer according to the second exemplary embodiment with a tray unit pulled out.
- (15) FIG. 15 is a diagram illustrating a printer according to a third exemplary embodiment with a door, a fixing device, and a cleaning unit moved.
- (16) FIG. 16 is a diagram illustrating the printer according to the third exemplary embodiment with a tray unit pulled out.
- (17) FIG. 17 illustrates the overall configuration of a printer according to a fourth exemplary embodiment.
- (18) FIG. 18 is a diagram illustrating the printer according to the fourth exemplary embodiment with a door and a transfer unit moved.
- (19) FIG. 19 is a diagram illustrating the printer according to the fourth exemplary embodiment with a tray unit pulled out.
- (20) FIG. 20 illustrates the overall configuration of a printer according to a fifth exemplary embodiment.
- (21) FIG. 21 is a diagram illustrating the printer according to the fifth exemplary embodiment with a door and a fixing device moved.
- (22) FIG. 22 is a diagram illustrating the printer according to the fifth exemplary embodiment with a tray unit pulled out.
- (23) FIG. 23 illustrates the overall configuration of a printer according to a sixth exemplary embodiment.
- (24) FIG. 24 is an overall view with a door, a transfer unit, and a conveyance unit moved according to the sixth exemplary embodiment.
- (25) FIG. 25 is an overall view of the sixth exemplary embodiment with a tray unit pulled out.

DESCRIPTION OF THE EMBODIMENTS

(26) Exemplary embodiments of the present disclosure will be described in detail below with reference to the drawings. It should be noted that dimensions, materials, shapes, and relative arrangement of components described in the exemplary embodiments are subject to appropriate changes depending on the configurations of the apparatuses to which the exemplary embodiments are applied and various conditions. In other words, the scope of the disclosure is not intended to be limited to the following exemplary embodiments.

(27) [Overall Configuration]

(28) A first exemplary embodiment will be described. A printer **1** that is an image forming apparatus will be described with reference to FIG. **1**. FIG. **1** illustrates an overall configuration of the printer **1** according to the present exemplary embodiment. In the present exemplary embodiment, the printer **1** is an electrophotographic color laser beam printer that forms an image on a sheet **S** serving as a recording material.

(29) The printer **1** includes an apparatus main body (housing) **1A**, a scanner (exposure device) **2**, a control unit **3**, and a door (opening and closing member) **20** openable and closable with respect to the apparatus main body **1A**. The printer **1** further includes a sheet feed unit **30**, a transfer unit (transfer device) **40**, a tray unit (moving unit or support unit) **50**, and a fixing device **80**. A portion including the apparatus main body **1A** and the door **20** may be referred to as a main frame **100**. The main frame **100** includes an exterior cover portion of the printer **1**.

(30) The apparatus main body **1A** accommodates the scanner **2**, the control unit **3**, the sheet feed unit **30**, the transfer unit **40**, the tray unit **50**, and the fixing device **80**.

(31) The sheet feed unit **30** includes a stacking tray **31** where sheets **S** serving as a recording material are stacked, and a feed roller **32**. The stacking tray **31** can be pulled out in a direction toward the door **20** and replenished with sheets **S**.

(32) The tray unit **50** includes a tray (support member or drawer) **51** and cartridges **PY**, **PM**, **PC**, and **PK**. The tray **51** includes a tray handle **52**. The cartridges **PY**, **PM**, **PC**, and **PK** are detachably attached to the tray **51**.

(33) The cartridges **PY**, **PM**, **PC**, and **PK** can be attached to and detached from the tray **51** independent of each other. The cartridges **PY**, **PM**, **PC**, and **PK** accommodate yellow (**Y**), magenta (**M**), cyan (**C**), and black (**K**) toners (developers), respectively. The cartridges **PY**, **PM**, **PC**, and **PK** have the same configuration except that toners of different colors are accommodated. A configuration and operation of one of the cartridges **PY**, **PM**, **PC**, and **PK** will therefore be described, and a description of the others will be omitted in some cases. If the cartridges **PY**, **PM**, **PC**, and **PK** do not need to be distinguished, the cartridges **PY**, **PM**, **PC** and **PK** may be referred to simply as cartridges **P**. The tray unit **50** can be said to include a plurality of cartridges **P** and the tray **51** to which the plurality of cartridges **P** is detachably attached.

(34) In the present exemplary embodiment, the tray unit **50** includes a plurality of photosensitive drums (image bearing members) **61**, a plurality of charging rollers (charging members) **62**, and a plurality of developing rollers (developer bearing members) **71**. Specifically, the tray unit **50** includes four photosensitive drums **61**, four charging rollers **62**, and four developing rollers **71**. The directions of the rotation axes of the photosensitive drums **61**, those of the developing rollers **71**, and those of the charging rollers **62** are parallel.

(35) The portion to form a **K** image will be referred to as a **K** station (first station). The photosensitive drum **61** of the first station will be referred to as a first photosensitive drum, the developing roller **71** a first developing roller, and the charging roller **62** a first charging roller.

(36) The portion to form a **C** image will be referred to as a **C** station (second station). The photosensitive drum **61** of the second station will be referred to as a second photosensitive drum, the developing roller **71** a second developing roller, and the charging roller **62** a second charging roller.

(37) The portion to form an **M** image will be referred to as an **M** station (third station). The

photosensitive drum **61** of the third station will be referred to as a third photosensitive drum, the developing roller **71** a third developing roller, and the charging roller **62** a third charging roller.

(38) The portion to form a Y image will be referred to as a Y station (fourth station). The photosensitive drum **61** of the fourth station will be referred to as a fourth photosensitive drum, the developing roller **71** a fourth developing roller, and the charging roller **62** a fourth charging roller.

(39) The cartridge PK is attached to the K station, the cartridge PC to the C station, the cartridge PM to the M station, and the cartridge PY to the Y station. In the present exemplary embodiment, the cartridge PK will be referred to as a first cartridge, the cartridge PC a second cartridge, the cartridge PM a third cartridge, and the cartridge PY a fourth cartridge.

(40) The ordinal numbers the first, second, third, and fourth are used for the convenience of description.

(41) The photosensitive drums **61**, the charging rollers **62**, and the developing rollers **71** may be provided either in the cartridges P or on the tray **51**. In the present exemplary embodiment, the cartridges P include the photosensitive drums **61**, the charging rollers **62**, and the developing rollers **71**.

(42) The transfer unit **40** includes a belt **41**, primary transfer rollers **42**, a cleaning unit **43**, a driving roller **46** for driving the belt **41**, and a tension roller (driven roller) **47**. The printer **1** according to the present exemplary embodiment includes an optical sensor **44** for detecting a toner image transferred to the belt **41**. In the present exemplary embodiment, the belt **41** is disposed under the photosensitive drums **61**, and can contact the photosensitive drums **61** to form primary transfer portions between the belt **41** and the photosensitive drums **61**. The printer **1** also includes a secondary transfer roller **45** that contacts the belt **41** to form a secondary transfer portion. The secondary transfer portion is formed between the belt **41** and the secondary transfer roller **45**. The directions of the rotation axes of the primary transfer rollers **42**, that of the driving roller **46**, that of the tension roller **47**, and that of the secondary transfer roller **45** are parallel. A registration roller pair **4** is arranged in front of the secondary transfer portion.

(43) The fixing device **80** includes a fixing unit **81** and a flapper **5**. When an image forming operation is performed on a sheet S, the fixing device **80** is at a use position. The fixing device **80** is accommodated at the use position in (inside) the apparatus main body **1A**. The fixing device **80** is configured to heat the sheet S at the use position. In the present exemplary embodiment, the fixing unit **81** includes a heating unit (heating roller) including a heater, and a pressure unit (pressure roller).

(44) A movement of the transfer unit **40** and the tray unit **50** will be described with reference to FIGS. **1**, **2**, **3**, **4**, and **5**. FIG. **2** is a diagram illustrating the printer **1** with the door **20** open. FIG. **3** is a diagram illustrating the printer **1** with the fixing device **80** moved. FIG. **4** is a diagram illustrating the printer **1** with the transfer unit **40** and the tray unit **50** pulled out. FIG. **5** is a diagram illustrating the printer **1** with only the transfer unit **40** pulled out.

(45) The transfer unit **40** and the tray unit **50** can be moved from inside to outside the apparatus main body **1A**. The apparatus main body **1A** has a first end **1b1** provided with an opening (main body opening) **1A1** and a second end **1b2** opposite to the first end **1b1** in a horizontal direction H. The tray unit **50** can be moved to a first inner position inside the apparatus main body **1A** and to a first outer position outside the apparatus main body **1A** through the opening **1A1**. The transfer unit **40** can be moved to a second inner position inside the apparatus main body **1A** and to a second outer position outside the apparatus main body **1A** through the opening **1A1**. The opening **1A1** may include an opening for the tray unit **50** to pass through and an opening for the transfer unit **40** to pass through. When the transfer unit **40** is moved from the second inner position to the second outer position, at least the belt **41** is moved so that at least a part of the belt **41** protrudes outward from the apparatus main body **1A**.

(46) The direction in which the tray unit **50** is moved from the first inner position to the first outer positions will be referred to as a tray detachment direction Dd1. The direction opposite to the tray

detachment direction Dd1 will be referred to as a tray attachment direction Da1. The tray detachment direction Dd1 can be said to be a direction from the second end 1b2 to the first end 1b1.

(47) The direction in which the transfer unit 40 is moved from the second inner position to the second outer position will be referred to as a transfer detachment direction Dd2. The direction opposite to the transfer detachment direction Dd2 will be referred to as a transfer attachment direction Da2. The driving roller 46 is disposed downstream of the tension roller 47 in the transfer detachment direction Dd2. The transfer detachment direction Dd2 can be said to be a direction from the second end 1b2 to the first end 1b1.

(48) The tray detachment direction Dd1 and the tray attachment direction Da1 are directions intersecting with (desirably, orthogonal to) the directions of the rotation axes of the photosensitive drums 61. The transfer detachment direction Dd2 and the transfer attachment direction Da2 are directions intersecting with (desirably, orthogonal to) the direction of the rotation axis of the driving roller 46. The direction of the rotation axis of the driving roller 46 is parallel to those of the photosensitive drums 61.

(49) The fixing device 80 is disposed at one end side (side where the first end 1b1 is located) of the apparatus main body 1A in the horizontal direction H.

(50) The door 20 attached to the apparatus main body 1A can be moved between a closed position and an open position. As illustrated in FIG. 1, at the closed position (closed state of the door 20), the door 20 covers the opening 1A1. As illustrated in FIG. 2, at the open position (open state of the door 20), the opening 1A1 is exposed.

(51) As illustrated in FIG. 1, with the door 20 at the closed position, the door 20 covers the fixing device 80 attached to the apparatus main body 1A. More specifically, with the door 20 at the closed position, an upper cover portion 20b of the door 20 is located above the fixing device 80. The upper cover portion 20b of the door 20 has a function as a part of the exterior cover portion.

(52) The door 20 can be moved to the open position and the closed position with the fixing device 80 supported by the apparatus main body 1A. In other words, the door 20 is moved from the closed position to the open position as separating (getting away) from the fixing device 80 supported by the apparatus main body 1A. As illustrated in FIG. 2, when the door 20 is at the open position, the door 20 is away from the fixing device 80 supported by the main body 1A.

(53) As will be described below, the fixing device 80 can be moved from the state of FIG. 2 to the state of FIG. 3 so that the opening 1A1 is exposed wide. With the door 20 and the fixing device 80 moved (FIG. 3), the transfer unit 40 and the tray unit 50 can be moved from inside to outside the apparatus main body 1A through the opening 1A1. FIG. 4 illustrates the state after the movement.

(54) With the tray unit 50 moved out of the apparatus main body 1A (FIG. 4), the cartridges PY, PM, PC, and PK can be detached from the tray 51, and can be attached to the tray 51. The cartridges PY, PM, PC, and PK can thereby be replaced with new cartridges PY, PM, PC, and PK. In the present exemplary embodiment, the cartridges P can be attached to and detached from the tray 51 in a direction intersecting with (desirably, orthogonal to) the rotation axes of the photosensitive drums 61. The cartridges PY, PM, PC, and PK are detached from the tray 51 by being moved in a direction away from the transfer unit 40. In other words, the cartridges PY, PM, PC, and PK are detached from the tray 51 by being moved in a direction opposite from the transfer unit 40. In the present exemplary embodiment, the transfer unit 40 is disposed under the tray unit 50. The cartridges PY, PM, PC, and PK are therefore detached from the tray 51 by being moved up from the tray 51.

(55) Moreover, the transfer unit 40 can be detached from the apparatus main body 1A independent of the tray unit 50, and can be replaced with a new transfer unit 40.

(56) [Image Forming Operation]

(57) An image forming operation by the printer 1 will be described with reference to FIG. 1. The control unit 3 of the printer 1 starts an image forming operation on a sheet S based on an image

signal received from an external host apparatus **400**. Examples of the external host apparatus **400** include a personal computer, an image reader, and a facsimile.

(58) In performing image formation on the sheet **S**, the fixing device **80** is positioned at the use position, the tray unit **50** at the first inner position, the transfer unit **40** at the second inner position, and the door **20** at the closed position. With the transfer unit **40** at the second inner position, the belt **41** can contact the photosensitive drum **61**. Here, the tray unit **50** is located above the transfer unit **40**.

(59) A charging voltage is applied to the charging roller **62** as the photosensitive drum **61** is rotated. The scanner **2** irradiates the photosensitive drum **61** with laser corresponding to image information, whereby the surface of the photosensitive drum **61** charged by the charging roller **62** is exposed, forming an electrostatic latent image corresponding to the image information on the surface of the photosensitive drum **61**.

(60) The developing roller **71** bears toner. A developing voltage is applied to the developing roller **71**, and the electrostatic latent image formed on the photosensitive drum **61** is developed with toner supplied from the developing roller **71**, whereby a toner image is formed on the surface of the photosensitive drum **61**. In the present exemplary embodiment, the developing roller **71** in contact with the photosensitive drum **61** develops the electrostatic latent image. However, the developing roller **71** may develop the electrostatic latent image through a gap between the developing roller **71** and the photosensitive drum **61**.

(61) In forming a full-color image, toner images of respective colors are formed on the photosensitive drums **61**.

(62) In the present exemplary embodiment, with the tray unit **50** at the first inner position, the developing roller **71** can be moved between a contact position where the developing roller **71** contacts the photosensitive drum **61** and a separated position where the developing roller **71** is separated from the photosensitive drum **61**. Specifically, the state where the developing roller **71** is at the contact position and the state where the developing roller **71** is at the separated position are switched by a switching device included in the apparatus main body **1A**. The developing roller **71** can thus be kept separated from the photosensitive drum **61** in a state where no image forming operation is performed.

(63) The printer **1** can perform monochrome printing in a state where the developing roller **71** and the photosensitive drum **61** corresponding to the cartridge PK are in contact with each other and the developing rollers **71** and the photosensitive drums **61** corresponding to the cartridges PY, PM, and PC are separated from each other. The printer **1** can perform full-color printing in a state where the photosensitive drums **61** corresponding to the cartridges PY, PM, PC, and PK contact the belt **41**.

(64) The toner image formed on each photosensitive drum **61** is transferred to the belt **41** by the primary transfer roller **42** at the primary transfer portion, and conveyed to the secondary transfer portion formed by the belt **41** and the secondary transfer roller **45**.

(65) The apparatus main body **1A** includes a conveyance path (first path or first conveyance path) **1c** for the sheet **S** traveling toward the fixing device **80** to pass through. The door **20** includes a two-sided conveyance path (second path or second conveyance path) **20a** for the sheet **S** passed through the fixing device **80** to pass through. The door **20** in the closed state covers the conveyance path **1c**. With the door **20** open, the conveyance path **1c** and the two-sided conveyance path **20a** are exposed (FIG. 2).

(66) In the sheet feed unit **30**, each of the sheets **S** stacked in the stacking tray **31** is separated and fed by the feed roller **32** at predetermined timing, and conveyed toward the secondary transfer portion and the fixing device **80** through the conveyance path **1c**.

(67) At the secondary transfer portion, the toner image is transferred from the belt **41** to the sheet **S**. Toner not transferred to the sheet **S** is removed from the belt **41** with a cleaning blade (cleaning member) **43A** arranged in the cleaning unit **43**.

(68) The sheet **S** to which the toner image is transferred at the secondary transfer portion is

conveyed toward the fixing device **80**. In the fixing device **80**, the sheet S is heated and pressed by the fixing unit **81**, whereby the toner image is fixed to the sheet S. The sheet S to which the toner image is fixed is conveyed toward the flapper **5** serving as a path switching unit.

(69) The flapper **5** can be moved to a sheet discharge position where the sheet S that has passed through the fixing device **80** is guided to a sheet discharge path **1d** and a reversing position where the sheet S is guided to a reversing path **1e**.

(70) In performing one-sided printing where an image is formed on one side of the sheet S, the sheet S is guided to the sheet discharge path **1d** by the flapper **5**, and discharged to a discharge tray **1f** formed in the upper part of the apparatus main body **1A**.

(71) In performing two-sided printing where images are printed on the front and back of the sheet S, the sheet S is guided to the reversing path **1e** by the flapper **5**. After the sheet S is guided to the reversing path **1e**, the conveyance direction of the sheet S is reversed and the sheet S is conveyed toward the secondary transfer portion through the two-sided conveyance path **20a** formed in the door **20**. At the secondary transfer portion, a toner image is transferred to the back of the sheet S. The sheet S then passes through the fixing device **80**, is guided to the sheet discharge path **1d** by the flapper **5**, and discharged to the discharge tray **1f** of the apparatus main body **1A**.

(72) [Detachment of Tray Unit and Transfer Unit and Layout of Fixing Device]

(73) The detachment of the tray unit **50** and the transfer unit **40** from the apparatus main body **1A** and the layout of the fixing device **80** will be described with reference to FIGS. **1**, **2**, **3**, **4**, and **5**.

(74) As described above, the fixing device **80** of the printer **1** is disposed at one end side of the apparatus main body **1A** in the horizontal direction H. The transfer unit **40** and the tray unit **50** are moved out of the apparatus main body **1A** from the one end side of the apparatus main body **1A** through the opening **1A1**.

(75) In other words, in the horizontal direction H, the fixing device **80** is closer to the first end **1b1** than to the second end **1b2**. That is, in the horizontal direction H, the distance between the fixing device **80** and the first end **1b1** is smaller than the distance between the fixing device **80** and the second end **1b2**. The fixing device **80** can be said to be positioned closer to the first end **1b1** than the center of the apparatus main body **1A** is in the horizontal direction H. The distance between the first end **1b1** and the fixing device **80** in the horizontal direction H can be said to be smaller than the distance between the first end **1b1** and the center of the apparatus main body **1A**.

(76) As described above, when the tray unit **50** is moved from the first inner position to the first outer position, the moving direction of the tray unit **50** intersects with the rotation axes of the photosensitive drums **61**. Here, the tray unit **50** is moved away from the second end **1b2**.

(77) As described above, when the transfer unit **40** is moved from the second inner position to the second outer position, the moving direction of the transfer unit **40** intersects with the rotation axis of the driving roller **46**. Here, the transfer unit **40** is moved away from the second end **1b2**.

(78) As described above, the door **20** according to the present exemplary embodiment covers the opening **1A1** and at least a part of the conveyance path **1c** of the sheet S in the closed state.

Moreover, the door **20** includes the two-sided conveyance path **20a**.

(79) If the sheet S jams, the user of the printer **1** can clear the jam by removing the sheet S from the one end side of the apparatus main body **1A**. Specifically, the user can move the door **20** to the open position and access the interior of the apparatus main body **1A** to remove the sheet S. If a part of the sheet S that has passed through the fixing device **80** is exposed outside the apparatus main body **1A**, the user can remove the sheet S by pulling the sheet S from outside the apparatus main body **1A** without opening the door **20**.

(80) The user of the printer **1** can further perform operations, such as status check, maintenance, and replacement of the transfer unit **40** and the cartridges P by moving the transfer unit **40** and the tray unit **50** out of the apparatus main body **1A** from the one end side of the apparatus main body **1A**.

(81) In other words, while the printer **1** includes the fixing device **80** disposed at the one end side

of the apparatus main body **1A**, the transfer unit **40** and the tray unit **50** can be moved into and out of the apparatus main body **1A** through the one end side of the apparatus main body **1A**. As a result, the user can clear a jam, access the fixing device **80**, and operate the transfer unit **40** and the tray unit **50** in one direction.

(82) In the present exemplary embodiment, the front of the printer **1** is the door **20** side. A working space for clearing a jam and operating the transfer unit **40** and the tray unit **50** can thus be provided in front of the printer **1**. By contrast, a space for performing such operations from the left, right, rear, or top of the printer **1** does not need to be provided. The printer **1** can thus be installed in a space saving manner.

(83) Suppose that the transfer unit **40** or the tray unit **50** is pulled out from the other end side of the apparatus main body **1A**. In such a case, to clear a jam and pull out the transfer unit **40** or the tray unit **50**, the user accesses the apparatus main body **1A** from both sides. If the transfer unit **40** or the tray unit **50** is pulled out from one end side of the apparatus main body **1A** and the other from the other end side, the user also accesses the apparatus main body **1A** from both sides. The area for installing the printer **1** thus increases due to the presence of working spaces behind the apparatus main body **1A**, as well as in front of the apparatus main body **1A**.

(84) Moreover, in the present exemplary embodiment, the apparatus main body **1A** can be replenished with sheets **S** from the one end side. The space for replenishing the printer **1** with sheets **S** can thus be provided in front, and the printer **1** can be installed in a space saving manner.

(85) [Relationship Between Tray Unit and Fixing Device]

(86) A relationship between the tray unit **50** and the fixing device **80** will be described with reference to FIGS. **1**, **2**, **3**, **4**, **6**, and **7**. FIGS. **6** and **7** illustrate the movement of the transfer unit **40** and the tray unit **50**.

(87) The tray unit **50** passes through a predetermined space when being moved between the first inner position where the image forming operation is performed on a sheet **S** and the first outer position where the cartridges **P** can be replaced.

(88) The space for the tray unit **50** to pass through in being moved from the first inner position to the first outer position will be referred to as a first space (moving space of the tray unit **50**). The first space can also be referred to as a path or locus (first moving path or first moving locus) for the tray unit **50** to pass through in being moved from the first inner position to the first outer position.

(89) The transfer unit **40** passes through a predetermined space when being moved between the second inner position where the image forming operation is performed on a sheet **S** and the second outer position. The space for the transfer unit **40** to pass through in being moved from the second inner position to the second outer position will be referred to as a second space (moving space of the transfer unit **40**). The second space can also be referred to as a path or locus (second moving path or second moving locus) for the transfer unit **40** to pass through in being moved from the second inner position to the second outer position.

(90) The position of the fixing device **80** in performing the image forming operation on a sheet **S** (use position), the first space, and the second space have the following relationship.

(91) As illustrated in FIGS. **6** and **7**, at least a part of the fixing device **80** at the use position overlaps with the first space. In other words, with the fixing device **80** at the use position, at least a part of the fixing device **80** is located in the first space.

(92) In the present exemplary embodiment, at least a part of the fixing device **80** overlaps with the space for the cartridges **P** to pass through when the tray unit **50** is moved from the first inner position to the first outer position. However, at least a part of the fixing device **80** may be positioned to overlap with the space for the tray **51** to pass through.

(93) This enables the miniaturization of the apparatus main body **1A** as compared to a configuration where the fixing device **80** at the use position is located outside the first space.

(94) As illustrated in FIG. **6**, in the present exemplary embodiment, at least a part of the two-sided conveyance path **20a** overlaps with the first space when the door **20** is in the closed state. In other

words, with the door **20** in the closed state, at least a part of the two-sided conveyance path **20a** is located in the first space.

(95) As described above, at least a part of the fixing device **80** at the use position is located in the first space. With the fixing device **80** at the use position, the movement of the tray unit **50** from the first inner position to the first outer position is thus restricted. Meanwhile, the fixing device **80** is configured to be retractable from the use position. With the fixing device **80** retracted from the use position, the tray unit **50** can be moved from the first inner position to the first outer position. When the tray unit **50** is moved from the first inner position to the first outer position, the tray unit **50** passes under the fixing device **80**. The fixing device **80**, whether at the use position or the retracted position, is located outside the second space. In both the state where the fixing device **80** is at the use state and the state where the fixing device **80** is at the retracted position, the transfer unit **40** can thus be moved from the second inner position to the second outer position. When the transfer unit **40** is moved from the second inner position to the second outer position, the transfer unit **40** passes under the fixing device **80**.

(96) The fixing device **80** includes a fixing frame **80a** that supports the fixing unit **81**. When the fixing device **80** is retracted from the use position, the fixing frame **80a** is displaced with respect to the apparatus main body **1A** while supporting the fixing unit **81**.

(97) In pulling the tray unit **50** out of the apparatus main body **1A**, the door **20** is initially moved to the open position as illustrated in FIG. 2. In the present exemplary embodiment, with the fixing device **80** at the use position, the door **20** can be moved to the open position and the closed position. With the door **20** at the open position, the fixing device **80** is moved from the use position to the retracted position to exit the first space as illustrated in FIG. 3.

(98) In the printer **1** according to the present exemplary embodiment, the fixing device **80** can be moved between the use position and the retracted position retracted from the use position while attached to the apparatus main body **1A**. In the present exemplary embodiment, with the fixing device **80** at the retracted position, the entire fixing device **80** is located outside the first space. The fixing device **80** is lifted from the use position to the retracted position above the use position. In other words, the retracted position is a position higher than the use position. As illustrated in FIG. 7, with the fixing device **80** at the retracted position, at least a part of the fixing device **80** is positioned at a position protruding outward from the apparatus main body **1A**. Moreover, with the fixing device **80** at the retracted position, the position of at least a part of the fixing device **80** is higher than that of the upper cover portion **20b** of the door **20** at the closed position. This can maintain the size of the printer **1** in a vertical direction **V** with the fixing device **80** at the use position small while the movement of the fixing device **80** to the retracted position provides a space for the tray unit **50** to move. With the fixing device **80** at the retracted position, the door **20** is prevented from taking the closed position.

(99) Specifically, the printer **1** includes a coupling member (fixing coupling member, fixing link, or arm) **85** movably coupled to the apparatus main body **1A**. The fixing device **80** is coupled to the apparatus main body **1A** via the coupling member **85**.

(100) The coupling member **85** can be rotated about a rotation center **85A**. In pulling the tray unit **50** out of the apparatus main body **1A**, the coupling member **85** is moved out of the first space. The fixing device **80** is rotatably connected to one end of the coupling member **85**. The other end of the coupling member **85** is connected to the apparatus main body **1A** rotatably about the rotation center **85A**. The fixing device **80** can be moved from the use position to the retracted position and from the retracted position to the use position while being supported by the coupling member **85** (state of being coupled to the apparatus main body **1A** via the coupling member **85**). The coupling member **85** is configured to be movable (swingable) with respect to the apparatus main body **1A** and the fixing device **80**. In other words, the fixing device **80** is connected to the apparatus main body **1A** via the coupling member **85**. This can increase the amount of movement of the fixing device **80** when the fixing device **80** is moved between the retracted position and the use position, as

compared to a configuration where the fixing device **80** is directly connected to the apparatus main body **1A**.

(101) As illustrated in FIG. 7, when the tray unit **50** is moved from the first inner position to the first outer position, the angle formed between the moving direction of the tray unit **50** and the horizontal direction H is smaller than that formed between the moving direction of the tray unit **50** and the vertical direction V. Moreover, when the fixing device **80** is moved from the use position to the retracted position, the amount of movement of the fixing device **80** in the vertical direction V is greater than that in the horizontal direction H. In other words, when the fixing device **80** is moved from the use position to the retracted position, the amount of movement of the fixing device **80** in the direction perpendicular to the moving direction of the tray unit **50** is greater than that in the direction parallel to the moving direction of the tray unit **50**.

(102) Moreover, the fixing device **80** may be configured to be moved to the use position and the retracted position while being attached to the apparatus main body **1A** via a guide fixed to the apparatus main body **1A**. In such a case, the guide may have any given shape.

(103) The method for moving the fixing device **80** out of the first space is not limited thereto. For example, the fixing device **80** may be configured to be detachable from the apparatus main body **1A** when retracted from the use position, and the fixing device **80** may thereby be moved out of the first space. In other words, the fixing device **80** may be positioned outside the first space by separating the fixing device **80** from the apparatus main body **1A** and the door **20** (separating the fixing device **80** from the main frame **100**). Alternatively, the fixing device **80** may be coupled to the door **20** so that the fixing device **80** is moved out of the first space by opening the door **20**.

(104) [Relationship Between Tray Unit and Transfer Unit]

(105) A relationship between the tray unit **50** and the transfer unit **40** will be described with reference to FIGS. 1, 2, 3, 4, 5, 6, and 7.

(106) As illustrated in FIG. 3, with the fixing device **80** positioned outside the first space, the user grips the tray handle **52** and moves the tray unit **50** to outside the apparatus main body **1A**.

(107) As illustrated in FIGS. 6 and 7, the position of the transfer unit **40** in performing the image forming operation on the sheet S and the first space have the following relationship.

(108) In performing the image forming operation on the sheet S, the transfer unit **40** is at the second inner position. As illustrated in FIGS. 6 and 7, at least a part of the transfer unit **40** at the second inner position overlaps with the first space. In other words, with the transfer unit **40** at the second inner position, at least a part of the transfer unit **40** is located in the first space.

(109) In the present exemplary embodiment, at least a part of the transfer unit **40** overlaps with the space for the tray **51** to pass through when the tray unit **50** is moved from the first inner position to the first outer position. However, at least a part of the transfer unit **40** may be positioned overlapping with a space for the cartridges P to pass through.

(110) The apparatus main body **1A** can thus be reduced in size as compared to a configuration where the transfer unit **40** at the second inner position is located outside the first space.

(111) In the printer **1** according to the present exemplary embodiment, the part of the transfer unit **40** overlapping with the first space refers to the cleaning unit **43** accommodating the cleaning member **43A**. However, other parts of the transfer unit **40** may overlap with the first space. In the present exemplary embodiment, the cleaning member **43A** of the transfer unit **40** at the second inner position overlaps with the first space.

(112) The tray unit **50** and the transfer unit **40** can be said to satisfy the following relationship.

(113) In the state where the tray unit **50** is at the first inner position and the transfer unit **40** is at the second inner position, a part of the transfer unit **40** (cleaning unit **43**) is located downstream of the tray unit **50** in the moving direction where the tray unit **50** is moved from the first inner position to the first outer position. In the vertical direction V, an area (range) where the part of the transfer unit **40** (cleaning unit **43**) downstream of the tray unit **50** is positioned overlaps at least in part with an area (range) where the tray unit **50** is positioned.

(114) In the present exemplary embodiment, when the door **20** is in the closed state, at least a part of the two-sided conveyance path **20a** overlaps with the second space for the transfer unit **40** to pass through in being moved from the second inner position to the second outer position. In other words, when the door **20** is in the closed state, at least a part of the two-sided conveyance path **20a** is located in the second space.

(115) As described, with the transfer unit **40** at the second inner position, at least a part of the transfer unit **40** is located in the first space. In the present exemplary embodiment, when the user moves the tray unit **50** with respect to the apparatus main body **1A** from the first inner position to the first outer position, the transfer unit **40** is thus also moved with respect to the apparatus main body **1A** from the second inner position to the second outer position. In other words, the tray unit **50** is moved from the first inner position to the first outer position and the transfer unit **40** from the second inner position to the second outer position in a state where the transfer unit **40** and the tray unit **50** are integrated with each other.

(116) The integral movement of the tray unit **50** and the transfer unit **40** can prevent the photosensitive drums **61** from being exposed when the tray unit **50** is moved to the first outer position, and can prevent damage or stain to the photosensitive drums **61**.

(117) As illustrated in FIG. 7, with the door **20** open, the tray unit **50** at the first outer position, and the transfer unit **40** at the second outer position, the transfer unit **40** supports the tray unit **50** from below the tray unit **50**. Moreover, in the present exemplary embodiment, the door **20** supports the transfer unit **40**. In other words, the transfer unit **40** supported by the door **20** supports the tray unit **50**. In such a state, the door **20** can be said to support the tray unit **50** via the transfer unit **40**.

(118) A configuration for integrally moving the tray unit **50** and the transfer unit **40** will be described with reference to FIGS. 8, 9, and 10. FIGS. 8, 9, and 10 illustrate a coupling member (lever, stopper, or lock member) **53** that couples the tray **51** of the tray unit **50** with the transfer unit **40**.

(119) The printer **1** according to the present exemplary embodiment includes the lock member **53**. In the present exemplary embodiment, the lock member **53** is disposed on the tray **51** of the tray unit **50**. The lock member **53** includes a transfer lock portion **53A**, and can be rotated about a rotation center **53B**.

(120) The tray **51** includes a transfer coupling portion **51A**. The transfer unit **40** includes a transfer frame **48** that supports the driving roller **46**, the tension roller **47**, and the primary transfer rollers **42**. The transfer frame **48** includes a tray coupling groove **48A** and a lock coupling portion **48B**.

(121) The lock member **53** can be rotated about the rotation center **53B** into a coupling position where the transfer lock portion **53A** is engaged with the lock coupling portion **48B** and the transfer unit **40** is coupled with the tray unit **50**, and into a release position retracted from the coupling position.

(122) The tray coupling groove **48A** is engaged with the transfer coupling portion **51A**, whereby the movement of the tray unit **50** with respect to the transfer unit **40** in the first detachment direction **Dd1** and a direction orthogonal to the first detachment direction **Dd1** is restricted. Meanwhile, the movement of the transfer unit **40** with respect to the tray unit **50** in the second attachment direction **Da2** and a direction orthogonal to the second attachment direction **Da2** is restricted.

(123) Similar configurations are provided at a total of four locations, namely, near both ends of the tension roller **47** in the direction of the rotation axis of the tension roller **47** and near both ends of the driving roller **46** in the direction of the rotation axis of the driving roller **46**.

(124) The engagement of the transfer lock portion **53A** of the lock member **53** at the coupling position with the lock coupling portion **48B** restricts the movement of the tray unit **50** with respect to the transfer unit **40** in the first detachment direction **Da1**. Meanwhile, the movement of the transfer unit **40** with respect to the tray unit **50** in the second detachment direction **Dd2** is restricted.

(125) The lock member **53** and the lock coupling portion **48B** may also be disposed at both ends of

the transfer unit **40** in the direction of the rotation axis of the tension roller **47**.

(126) In the state where the transfer coupling portion **51A** is engaged with the tray coupling groove **48A** and the lock member **53** at the coupling position is engaged with the lock coupling portion **48B**, the movement of the transfer unit **40** and the tray unit **50** with respect to each other is restricted. In this state, the transfer unit **40** and the tray unit **50** may have some play.

(127) In other words, the transfer coupling portion **51A** and the lock member **53** can be said to have a function as a tray-side coupling device, and the tray coupling groove **48A** and the lock coupling portion **48B** have a function as a transfer-side coupling device. The coupling of the tray- and transfer-side coupling devices couples the tray unit **50** with the transfer unit **40**, and the tray unit **50** and the transfer unit **40** are moved integrally.

(128) As illustrated in FIG. **10**, the apparatus main body **1A** includes an unlocking portion **1B**. In the state where the tray unit **50** is at the first inner position and the transfer unit **40** is at the second inner position, the lock member **53** is in contact with the unlocking portion **1B** and positioned at an unlocked position. In such a state, the transfer lock portion **53A** is separated from the lock coupling portion **48B**.

(129) If the tray unit **50** is moved in the first detachment direction **Dd1** with the lock member **53** at the unlocked position, the transfer unit **40** is pushed in the second detachment direction **Dd2** by the transfer coupling portion **51A**. As a result, the transfer unit **40** is also moved in the second detachment direction **Dd2** integrally with the tray unit **50**. If the tray unit **50** is moved in the first detachment direction **Dd1** by a predetermined distance, the lock member **53** gets separated from the unlocking portion **1B** and is moved to the coupling position, and is engaged with the lock coupling portion **48B**. The tray unit **50** is moved to the first outer position and the transfer unit **40** to the second outer position with the lock member **53** positioned at the coupling position and engaged with the lock coupling portion **48B**.

(130) If the tray unit **50** is moved in the first attachment direction **Da1** with the lock member **53** engaged with the lock coupling portion **48B**, the transfer unit **40** is pushed in the second attachment direction **Da2** by the lock member **53**. As a result, the transfer unit **40** is also moved in the second attachment direction **Da2** integrally with the tray unit **50**.

(131) If the transfer unit **40** is moved in the second attachment direction **Da2** with the lock member **53** engaged with the lock coupling portion **48B**, the transfer unit **40** pushes the transfer coupling portion **51A**, and the tray unit **50** is pushed in the first attachment direction **Da1**. As a result, the tray unit **50** is also moved in the first attachment direction **Da1** integrally with the transfer unit **40**.

(132) If the tray unit **50** and the transfer unit **40** are integrally moved into the apparatus main body **1A**, the lock member **53** is moved to the unlocked position by the unlocking portion **1B**. If the tray unit **50** is moved in the first attachment direction **Da1** in such a state, the transfer unit **40** is not moved. Then, as the door **20** is being closed, the transfer unit **40** is fully pushed into the apparatus main body **1A** (up to the second inner position) in the second attachment direction **Da2** by the contact force of the secondary transfer roller **45** with the transfer unit **40**. In other words, the door **20** can push the transfer unit **40** so that the transfer unit **40** is positioned at the second inner position. The door **20** may also push the tray unit **50** so that the tray unit **50** is positioned at the first inner position.

(133) Now, if the transfer unit **40** is moved in the second detachment direction **Dd2** with the lock member **53** at the unlocked position and the tray unit **50** at the first inner position, the transfer unit **40** can be moved from the second inner position to the second outer position independent of the tray unit **50**. In the printer **1** according to the present exemplary embodiment, as illustrated in FIG. **5**, the transfer unit **40** can be moved out of the apparatus main body **1A** (to the second outer position) with the tray unit **50** positioned inside the apparatus main body **1A** (at the first inner position). The transfer unit **40** can be detached from the apparatus main body **1A** and replaced with a new transfer unit **40**.

(134) In the present exemplary embodiment, in a state where the tray unit **50** is at the first inner

position, the tray unit **50** is positioned to the apparatus main body **1A** independent of the transfer unit **40**. In a state where the transfer unit **40** is at the second inner position, the transfer unit **40** is positioned to the apparatus main body **1A** independent of the tray unit **50**. The purpose is to give priority to accurate positioning of the transfer unit **40** and the tray unit **50** with respect to the apparatus main body **1A**. In the state where the tray unit **50** is at the first inner position and the transfer unit **40** is at the second inner position, there is therefore some clearance between the tray coupling groove **48A** and the transfer coupling portion **51A**.

(135) The clearance between the tray coupling groove **48A** and the transfer coupling portion **51A** may be eliminated so that the tray unit **50** and the transfer unit **40** can be positioned to each other in the state where the tray unit **50** is at the first inner position and the transfer unit **40** is at the second inner position. In such a case, the transfer unit **40** and the tray unit **50** can be accurately positioned to each other.

(136) [Movement of Tray Unit and Transfer Unit]

(137) As illustrated in FIGS. **1**, **6**, and **7**, the tray unit **50** and the transfer unit **40** while being attached to the apparatus main body **1A** are tilted with respect to the horizontal direction **H**.

(138) More specifically, with the tray unit **50** at the first inner position, the rotation axes of photosensitive drums **61** closer to the first end **1b1** is located, in the vertical direction **V**, above those of photosensitive drums **61** farther from the first end **1b1**.

(139) In other words, with the tray unit **50** at the first inner position, the second photosensitive drum is closer to the first end **1b1** than the first photosensitive drum is, and the rotation axis of the second photosensitive drum is located above that of the first photosensitive drum in the vertical direction **V**. Similarly, the third photosensitive drum is closer to the first end **1b1** than the second photosensitive drum is, and the rotation axis of the third photosensitive drum is located above that of the second photosensitive drum in the vertical direction **V**. The fourth photosensitive drum is closer to the first end **1b1** than the third photosensitive drum is, and the rotation axis of the fourth photosensitive drum is located above that of the third photosensitive drum in the vertical direction **V**.

(140) A straight line connecting the centers of the photosensitive drums **61** will be referred to as a line **LD**. In the present exemplary embodiment, when the tray unit **50** is at the first inner position, the line **LD** is tilted upward with respect to the horizontal direction **H** from the side farther from the first end position **1b1** to the side closer to the first end **1b1**.

(141) Moreover, the contact surface of the belt **41** to contact the first, second, third, and fourth photosensitive drums is tilted upward from the side farther from the first end **1b1** to the side closer to the first end **1b1**.

(142) When the tray unit **50** is moved from the first inner position to the first outer position, the tray unit **50** is moved in a direction oblique to the horizontal direction **H** (in the present exemplary embodiment, a direction tilted upward with respect to the horizontal direction **H**). In other words, when the tray unit **50** is moved from the first inner position to the first outer position, the tray unit **50** is moved obliquely upward.

(143) When the tray unit **50** is moved from the first inner position to the first outer position, the moving direction of the tray unit **50** lowers from obliquely upward. The moving direction of the tray unit **50** may include a direction parallel to the horizontal direction **H** or obliquely downward with respect to the horizontal direction **H**.

(144) The angle formed between the line **LD** and the horizontal direction **H** is smaller when the tray unit **50** is at the first outer position than when the tray unit **50** is at the first inner position. In the present exemplary embodiment, when the tray unit **50** is at the first outer position, the line **LD** and the horizontal direction **H** are parallel. However, the line **LD** may be tilted downward or upward with respect to the horizontal direction **H**.

(145) Similarly, when the transfer unit **40** is moved from the second inner position to the second outer position, the moving direction of the transfer unit **40** lowers from obliquely upward. The

moving direction of the transfer unit **40** may include a direction parallel to the horizontal direction H or obliquely downward with respect to the horizontal direction H.

(146) The angle formed between the contact surface of the belt **41** and the horizontal direction H is smaller when the transfer unit **40** is at the second outer position than when the transfer unit **40** is at the second inner position. In the present exemplary embodiment, when the transfer unit **40** is at the second outer position, the contact surface of the belt **41** and the horizontal direction H are parallel. However, the contact surface of the belt **41** may be tilted downward or upward with respect to the horizontal direction H.

(147) If the tray unit **50** is moved from the first outer position to the first inner position, the tray unit **50** is moved in the horizontal direction H and then obliquely downward from the middle. If the transfer unit **40** is moved from the second outer position to the second inner position, the transfer unit **40** is moved in the horizontal direction H and then obliquely downward from the middle. According to such a configuration, the own weights of the tray unit **50** and the transfer unit **40** can be used in inserting the tray unit **50** and the transfer unit **40** into the apparatus main body **1A**.

(148) (Modifications)

(149) As described above, the photosensitive drums **61**, the charging rollers **62**, and the developing rollers **71** may be disposed on either the cartridges P or the tray **51**. A description thereof will be given with reference to FIGS. **11A** to **11D**. FIGS. **11A** to **11D** are diagrams illustrating the tray unit **50** according to modifications.

(150) For example, as illustrated in FIG. **11A**, the tray **51** may include the photosensitive drums **61** and the charging rollers **62**, and the cartridges P may include the developing rollers **71**.

(151) As illustrated in FIG. **11B**, the cartridges P may include drum cartridges P1 including the photosensitive drums **61** and the charging rollers **62**, and developing cartridges P2 including the developing rollers **71**. In other words, the cartridges P may be separable into the drum cartridges P1 and the developing cartridges P2. In such a case, the drum cartridges P1 and the developing cartridges P2 can be detachably attachable to the tray **51** independent of each other.

(152) As illustrated in FIG. **11C**, the cartridges P may include cartridges PR including the photosensitive drums **61**, the charging rollers **62**, and the developing rollers **71**, and cartridges PT including toners to be replenished into the cartridges PR. In other words, the cartridges P may be separable into the cartridges PR and the cartridges PT.

(153) As illustrated in FIG. **11D**, the cartridges P may include drum cartridges P1 including the photosensitive drums **61** and the charging rollers **62**, developing cartridges P2 including the developing rollers **71**, and cartridges PT including toners to be replenished into the developing cartridges P2. In other words, the cartridges P may be separable into the drum cartridges P1, the developing cartridges P2, and the cartridges PT.

(154) A second exemplary embodiment will be described. In the second exemplary embodiment, components similar to those described in the foregoing exemplary embodiment are denoted by the same reference numerals as used in the exemplary embodiment. A description of the components similar to those of the exemplary embodiment may be omitted.

(155) [Relationship Between Tray Unit and Transfer Unit]

(156) A printer **101** that is an image forming apparatus according to the present exemplary embodiment includes an apparatus main body **101A** provided with an opening **101A1**. The printer **101**, the apparatus main body **101A**, and the opening **101A1** correspond to the printer **1**, the apparatus main body **1A**, and the opening **1A1** of the first exemplary embodiment, respectively.

(157) In the present exemplary embodiment, as in the first exemplary embodiment, with the transfer unit **40** at the second inner position, at least a part of the transfer unit **40** is located in the first space for the tray unit **50** to pass through in being moved from the first inner position to the first outer position. In the first exemplary embodiment, the tray unit **50** is configured to be moved out of the apparatus main body **1A** along with the transfer unit **40**. In the printer **101** according to the present exemplary embodiment, the transfer unit **40** can be moved so that a part of the transfer

unit **40** (specifically, the cleaning unit **43**) is positioned outside the first space. With the part of the transfer unit **40** retracted from the first space, the tray unit **50** is moved out of the apparatus main body **101A**.

(158) A relationship between the tray unit **50** and the transfer unit **40** will be described with reference to FIGS. **12**, **13**, and **14**. FIG. **12** illustrates an overall configuration of the printer **101** according to the second exemplary embodiment. FIG. **13** is a diagram illustrating the printer **101** with the door **20**, the fixing device **80**, and the transfer unit **40** moved. FIG. **14** is a diagram illustrating the printer **101** with the tray unit **50** pulled out.

(159) As in the first exemplary embodiment, before the tray unit **50** is moved, the door **20** and the fixing device **80** are retracted from the first space. Specifically, after the door **20** is moved to the open position, the fixing device **80** is moved from the use position to the retracted position. In the present exemplary embodiment, the transfer unit **40** is further tilted away from the tray unit **50** as illustrated in FIG. **13**. As a result, the cleaning unit **43** that is a part of the transfer unit **43** is moved out of the first space. In such a state, the belt **41** is separated from the photosensitive drums **61**.

(160) The user then grips the tray handle **52** and moves the tray unit **50** to outside the apparatus main body **101A**, whereby the tray unit **50** as illustrated in FIG. **14** can be pulled out of the apparatus main body **101A** through the opening **101A1**.

(161) According to the configuration of the present exemplary embodiment, the tray unit **50** can be moved out of the apparatus main body **101A** (to the first outer position) with the transfer unit **40** positioned inside the apparatus main body **101A**. Moreover, the transfer unit **40** can be moved to the second outer position with the tray unit **50** at the first inner position. The tray unit **50** can thus be moved by a small force as compared with a configuration where the tray unit **50** is moved out of the apparatus main body **101A** along with the transfer unit **40**. Again, in the present exemplary embodiment, the tray unit **50** passes under the fixing device **80** when being moved from the first inner position to the first outer position. The transfer unit **40** passes under the fixing device **80** when being moved from the second inner position to the second outer position.

(162) A third exemplary embodiment will be described. In the third exemplary embodiment, components similar to those described in the foregoing exemplary embodiments are denoted by the same reference numerals as used in the exemplary embodiments. A description of the components similar to those of the exemplary embodiments may be omitted.

(163) [Relationship Between Tray Unit and Transfer Unit]

(164) A printer **201** that is an image forming apparatus according to the present exemplary embodiment includes an apparatus main body **201A** provided with an opening **201A1**. The printer **201**, the apparatus main body **201A**, and the opening **201A1** correspond to the printer **1**, the apparatus main body **1A**, and the opening **1A1** of the first exemplary embodiment, respectively.

(165) In the present exemplary embodiment, as in the first and second exemplary embodiments, with the transfer unit **40** at the second inner position, at least a part of the transfer unit **40** is located in the first space for the tray unit **50** to pass through in being moved from the first inner position to the first outer position.

(166) In the printer **201** of the present exemplary embodiment, a part of the transfer unit **40** (cleaning unit **43**) can be moved with respect to the belt **41** so that the part of the transfer unit **40** (cleaning unit **43**) is positioned outside the first space for the tray unit **50** to pass through in being moved from the first inner position to the first outer position.

(167) A relationship between the tray unit **50** and the transfer unit **40** will be described with reference to FIGS. **15** and **16**. FIG. **15** is a diagram illustrating the printer **201** with the door **20**, the fixing device **80**, and the cleaning unit **43** moved. FIG. **16** is a diagram illustrating the printer **201** with the tray unit **50** pulled out.

(168) As in the first exemplary embodiment, before the tray unit **50** is moved, the door **20** and the fixing device **80** are retracted from the first space. In this third exemplary embodiment, the cleaning unit **43** is further rotated about the driving roller **46** as illustrated in FIG. **15**, whereby the

cleaning unit **43** is separated from the tray unit **50** and moved out of the first space.

(169) The cleaning unit **43** is moved with respect to the belt **41** out of the first space. The rotation of the cleaning unit **43** about the driving roller **46** can move the cleaning unit **43** without separating the cleaning blade **43A** from the belt **41**.

(170) The user then grips the tray handle **52** and moves the tray unit **50** to outside the apparatus main body **201A**, whereby the tray unit **50** can be pulled out of the apparatus main body **201A** as illustrated in FIG. **16**. While in the present exemplary embodiment the cleaning unit **43** is moved before the movement of the tray unit **50**, the cleaning unit **43** may be rotated with the movement of the tray unit **50**.

(171) According to the configuration of the present exemplary embodiment, the tray unit **50** can be moved out of the apparatus main body **201A** with the transfer unit **40** positioned inside the apparatus main body **201A**. The tray unit **50** can thus be moved by a small force as compared with a configuration where the tray unit **50** is moved out of the apparatus main body **201A** along with the transfer unit **40**. Again, in the present exemplary embodiment, the tray unit **50** passes under the fixing device **80** when being moved from the first inner position to the first outer position. The transfer unit **40** passes under the fixing device **80** when being moved from the second inner position to the second outer position.

(172) A fourth exemplary embodiment will be described with reference to FIGS. **17**, **18**, and **19**. In the fourth exemplary embodiment, components similar to those described in the foregoing exemplary embodiments are denoted by the same reference numerals as used in the exemplary embodiments. A description of the components similar to those of the exemplary embodiments may be omitted.

(173) A printer **301** that is an image forming apparatus according to the present exemplary embodiment includes an apparatus main body **301A** provided with an opening **301A1**. The printer **301**, the apparatus main body **301A**, and the opening **301A1** correspond to the printer **1**, the apparatus main body **1A**, and the opening **1A1** of the first exemplary embodiment, respectively.

(174) FIG. **17** illustrates an overall configuration of the printer **301** that is the image forming apparatus according to the present exemplary embodiment. FIG. **18** is an overall view of the printer **301** with the door **20** open. FIG. **19** is an overall view of the printer **301** with the tray unit **50** pulled out.

(175) Again, in the present exemplary embodiment, the tray unit **50** and the transfer unit **40** can be said to satisfy the following relationship. In the state where the tray unit **50** is at the first inner position and the transfer unit **40** is at the second inner position, a part of the transfer unit **40** (in the present exemplary embodiment, cleaning unit **43**) is located downstream of the tray unit **50** in the moving direction where the tray unit **50** is moved from the first inner position to the first outer position. In the vertical direction **V**, an area where the part of the transfer unit **40** (cleaning unit **43**) downstream of the tray unit **50** is positioned overlaps at least in part with an area where the tray unit **50** is positioned.

(176) The locus of movement of the tray unit **50** out of the apparatus main body **301A** according to the fourth exemplary embodiment will be described. As illustrated in FIG. **18**, the tray unit **50** is lifted away from the transfer unit **40** in an interlocked manner with the operation of opening the door **20**. The position of the tray unit **50** here may be referred to as a first intermediate position. If the door **20** is closed in such a state, the tray unit **50** is lowered to the first inner position. With the tray unit **50** at the first intermediate position, the belt **41** is separated from the photosensitive drums **61**.

(177) The tray unit **50** is then moved out of the apparatus main body **301A** through the opening **301A1** between the transfer unit **40** and the fixing device **80**.

(178) The printer **301** according to the fourth exemplary embodiment is configured so that, in the apparatus main body **301A**, the fixing device **80** at the use position and the transfer unit **40** at the second inner position are positioned outside the first space for the tray unit **50** to pass through in

being moved from the first inner position to the first outer position. The tray unit **50** can thus be moved out of the apparatus main body **301A** even with the fixing device **80** and the transfer unit **40** positioned at the same positions as when image formation is performed on a sheet **S**.

(179) While the apparatus main body **301A** according to the present exemplary embodiment entails a space for enabling the lifting of the tray unit **50**, the fixing device **80** and the transfer unit **40** do not need to be moved in pulling the tray unit **50** out of the apparatus main body **301A**. Moreover, the tray unit **50** can be moved by a small force as compared with a configuration where the tray unit **50** is moved out of the apparatus main body **301A** along with the transfer unit **40**. Again, in the present exemplary embodiment, the tray unit **50** passes under the fixing device **80** when being moved from the first inner position to the first outer position. The transfer unit **40** passes under the fixing device **80** when being moved from the second inner position to the second outer position.

(180) A fifth exemplary embodiment will be described with reference to FIGS. **20**, **21**, and **22**. In the fifth exemplary embodiment, components similar to those described in the foregoing exemplary embodiments are denoted by the same reference numerals as used in the exemplary embodiments. A description of the components similar to those of the exemplary embodiments may be omitted.

(181) A printer **401** that is an image forming apparatus according to the present exemplary embodiment includes an apparatus main body **401A** provided with an opening **401A1**, and a door **420**. The printer **401**, the apparatus main body **401A**, and the opening **401A1** correspond to the printer **1**, the apparatus main body **1A**, and the opening **1A1** of the first exemplary embodiment, respectively.

(182) FIG. **20** illustrates an overall configuration of the printer **401** that is the image forming apparatus according to the present exemplary embodiment. FIG. **21** is a diagram illustrating the printer **401** with the fixing device **80** moved. FIG. **22** is a diagram illustrating the printer **401** with the tray unit **50** pulled out.

(183) In the first to fourth exemplary embodiments, toner images formed on the photosensitive drums **61** are transferred to the belt **41** of the transfer unit **40**, and then transferred from the belt **41** to the sheet **S** at the secondary transfer portion. In other words, the transfer units **40** according to the first to fourth exemplary embodiments have a function as an intermediate transfer unit.

(184) By contrast, according to the fifth exemplary embodiment, a transfer unit (transfer device) **440** conveys a sheet **S** using a belt (conveying belt) **441** so that the sheet **S** contacts the photosensitive drums **61** and the toner images are transferred from the photosensitive drums **61** to the sheet **S**. In other words, the transfer unit **440** according to the present exemplary embodiment has a function as a conveyance belt unit.

(185) The printer **401** that is the image forming apparatus according to this fifth exemplary embodiment includes the apparatus main body **401A**, a sheet feed unit **430**, and the transfer unit **440**. As in the first exemplary embodiment, the fixing device **80** is disposed at one end side of the apparatus main body **401A**.

(186) The transfer unit **440** includes the belt **441** for conveying a sheet **S**, four primary transfer rollers **442**, a driving roller **446**, and a tension roller **447**. The printer **401** according to the present exemplary embodiment includes an optical sensor **44** for detecting a toner image transferred to the belt **441**.

(187) The tray unit **50** is moved out of the apparatus main body **401A** from the one end side of the apparatus main body **401A**. Specifically, the opening **401A1** for the tray unit **50** to pass through is arranged at the one end side of the apparatus main body **401A**. The door **420** can be moved between a closed position where the opening **401A1** is covered and an open position where the opening **401A1** is exposed.

(188) As in the first exemplary embodiment, toner images are formed on the photosensitive drums **61** based on image information. Meanwhile, a sheet **S** fed from the sheet feed unit **430** is conveyed in contact with the photosensitive drums **61** by the belt **441**. Here, transfer voltages are applied to the primary transfer rollers **442**, whereby the toner images are transferred from the photosensitive

drums **61** to the sheet **S**.

(189) The sheet **S** to which the toner images are transferred is conveyed to the fixing device **80**. The toner images are fixed to the sheet **S** by the fixing device **80**.

(190) In performing one-sided printing, the sheet **S** that has passed through the fixing device **80** is discharged by the flapper **5** to a discharge tray formed in the upper part of the apparatus main body **401A**. In performing two-sided printing, the sheet **S** reversed by the flapper **5** is delivered from a two-sided conveyance path **420a** inside the door **420** to the belt **441** again through a return conveyance path disposed above the sheet feed unit **430**. The sheet **S** is conveyed to the position to contact the photosensitive drums **61** by the belt **441**. After toner images are transferred from the photosensitive drums **61** to the back of the sheet **S**, the sheet **S** passes through the fixing device **80** and is discharged to the discharge tray.

(191) As in the first exemplary embodiment, in performing the image forming operation on the sheet **S**, at least a part of the fixing device **80** overlaps with the first space for the tray unit **50** to pass through in being moved from the first inner position to the first outer position. In moving the tray unit **50** out of the apparatus main body **401A**, the fixing device **80** is therefore moved out of the first space.

(192) Like the printer **1** according to the first exemplary embodiment, the printer **401** according to the present exemplary embodiment is also configured so that the transfer unit **440** and the tray unit **50** can be moved into and out of the apparatus main body **401A** through the one end side of the apparatus main body **401A** while the fixing device **80** is disposed at one end side of the apparatus main body **401A**. In other words, when the transfer unit **440** is moved from the second inner position to the second outer position, the transfer unit **440** is moved away from the other end side (second end **1b2**) of the apparatus main body **401A**. When the tray unit **50** is moved from the first inner position to the first outer position, the tray unit **50** is moved away from the other end side (second end **1b2**) of the apparatus main body **401A**. As a result, the user can work to clear a jam, access the fixing device **80**, and operate the transfer unit **440** and the tray unit **50** in one direction.

(193) In the present exemplary embodiment, the tray unit **50** can be singly moved to the first outer position. However, as in the first exemplary embodiment, the tray unit **50** may be moved to the first outer position along with the transfer unit **440**. Again, in the present exemplary embodiment, the tray unit **50** passes under the fixing device **80** when being moved from the first inner position to the first outer position. The transfer unit **440** passes under the fixing device **80** when being moved from the second inner position to the second outer position.

(194) A sixth exemplary embodiment will be described with reference to FIGS. **23**, **24**, and **25**. In the sixth exemplary embodiment, components similar to those described in the foregoing exemplary embodiments are denoted by the same reference numerals as used in the exemplary embodiments. A description of the components similar to those of the exemplary embodiments may be omitted.

(195) A printer **501** that is an image forming apparatus according to the present exemplary embodiment includes an apparatus main body **501A** provided with an opening **501A1**, and a door **20**. The printer **501**, the apparatus main body **501A**, and the opening **501A1** correspond to the printer **1**, the apparatus main body **1A**, and the opening **1A1** of the first exemplary embodiment, respectively.

(196) The printer **501** further includes a transfer unit (transfer device) **540**, a tray unit **550**, and a scanner **502**. The transfer unit **540**, the tray unit **550**, and the scanner **502** correspond to the transfer unit **40**, the tray unit **50**, and the scanner **2** of the first exemplary embodiment, respectively.

(197) In the printer **501** according to the present exemplary embodiment, the transfer unit **540** is disposed above the tray unit **550**, and the scanner **502** is disposed below the tray unit **550**.

(198) The printer **501** serving as an image forming apparatus will be described with reference to FIG. **23**. FIG. **23** illustrates an overall configuration of the printer **501**.

(199) The tray unit **550** according to the present exemplary embodiment includes a tray (support

member or drawer) **551** and cartridges PPY, PPM, PPC, and PPK. The tray **551** includes a tray handle **552**. The cartridges PPY, PPM, PPC, and PPK are detachably attached to the tray **551**. (200) The cartridges PPY, PPM, PPC, and PPK can be attached to and detached from the tray **551** independent of each other. The cartridges PPY, PPM, PPC, and PPK accommodate Y, M, C, and K toners (developers), respectively. The cartridges PPY, PPM, PPC, and PPK have the same configuration except that toners of different colors are accommodated. If the cartridges PPY, PPM, PPC, and PPK do not need to be distinguished, the cartridges PPY, PPM, PPC and PPK may therefore be referred to simply as cartridges PP.

(201) The tray unit **550** includes four photosensitive drums (image bearing members) **61**, four charging rollers (charging members) **62**, and four developing rollers (developer bearing members) **71**. In the present exemplary embodiment, the cartridge PPK will be referred to as a first cartridge, the cartridge PPC a second cartridge, the cartridge PPM a third cartridge, and the cartridge PPY a fourth cartridge. In the present exemplary embodiment, the cartridges PP include the photosensitive drums **61**, the charging rollers **62**, and the developing rollers **71**.

(202) The transfer unit **540** includes a belt **541**, primary transfer rollers **42**, a cleaning unit **43**, a driving roller **46** for driving the belt **541**, and a tension roller (driven roller) **47**.

(203) The printer **501** according to the present exemplary embodiment includes an optical sensor **44** for detecting a toner image transferred to the belt **541**, and a conveyance unit **590** including a conveyance roller **553**.

(204) The conveyance unit **590** is disposed upstream of a secondary transfer portion in the conveyance direction of the sheet S, and configured to convey the sheet S toward the secondary transfer portion and the fixing device **80**.

(205) In the present exemplary embodiment, the belt **541** is disposed over the photosensitive drums **61** and can contact the photosensitive drums **61** to form primary transfer portions. The printer **501** also includes a secondary transfer roller **45** that contacts the belt **541** to form the secondary transfer portion. The directions of the rotation axes of the primary transfer rollers **42**, that of the driving roller **46**, that of the tension roller **47**, and that of the secondary transfer roller **45** are parallel.

(206) FIG. **24** is a diagram illustrating the printer **501** with the door **20**, the transfer unit **540**, and the conveyance unit **590** moved. FIG. **25** is a diagram illustrating the printer **501** with the tray unit **550** pulled out.

(207) The transfer unit **540** and the tray unit **550** can be moved from inside to outside the apparatus main body **501A**. Specifically, the fixing device **80** is disposed at one end side of the apparatus main body **501A** in the horizontal direction. The opening **501A1** for the transfer unit **540** and the tray unit **550** to pass through is formed at the one end side.

(208) The door **20** attached to the apparatus main body **501A** can be moved between a closed position where the opening **501A1** is covered (FIG. **23**) and an open position (FIG. **24**).

(209) The apparatus main body **501A** includes a first end **501b1** having the opening **501A1** and a second end **501b2** opposite to the first end **501b1** in a horizontal direction H. The tray unit **550** can be moved to a first inner position inside the apparatus main body **501A** and a first outer position outside the apparatus main body **501A** through the opening **501A1**. The transfer unit **540** can be moved to a second inner position inside the apparatus main body **501A** and a second outer position outside apparatus main body **501A** through the opening **501A1**. The opening **501A1** may include an opening for the tray unit **550** to pass through and an opening for the transfer unit **540** to pass through.

(210) In the horizontal direction H, the fixing device **80** is closer to the first end **501b1** than to the second end **501b2**. In other words, in the horizontal direction H, the distance between the fixing device **80** and the first end **501b1** is smaller than the distance between the fixing device **80** and the second end **501b2**.

(211) When the tray unit **550** is moved from the first inner position to the first outer position, the moving direction of the tray unit **550** intersects with the rotation axes of the photosensitive drums

61. Here, the tray units **550** is moved away from the second end **501b2**.

(212) As described above, when the transfer unit **540** is moved from the second inner position to the second outer position, the moving direction of the transfer unit **540** intersects with the rotation axes of the photosensitive drums **61**. Here, the transfer unit **540** is moved away from the second end **501b2**.

(213) In the present exemplary embodiment, at least a part of the conveyance unit **590** at the position where the image forming operation is performed on the sheet **S** is located in a first space for tray unit **550** to pass through in being moved from the first inner position to the first outer position.

(214) In the present exemplary embodiment, in moving the tray unit **550** from the first inner position to the first outer position, the transfer unit **540** is tilted away from the tray unit **550** as illustrated in FIG. **24**.

(215) Meanwhile, the conveyance unit **590** is rotated clockwise to move away from the tray unit **550** and is moved out of the first space. The user then grips the tray handle **552** and moves the tray unit **550** to outside the apparatus main body **501A**, whereby the tray unit **550** as illustrated in FIG. **25** can be pulled out of the apparatus main body **501A**.

(216) With the tray unit **550** moved out of the apparatus main body **501A** (FIG. **25**), the cartridges PPY, PPM, PPC, and PPK can be detached from the tray **551**, and can be attached to the tray **551**. In such a manner, the cartridges PPY, PPM, PPC, and PPK can be replaced with new cartridges PPY, PPM, PPC, and PPK.

(217) The transfer unit **540** can further be detached from the apparatus main body **501A** and replaced with a new transfer unit **540**.

(218) In the printer **501** according to the present exemplary embodiment, like the printer **1** of the first exemplary embodiment, the transfer unit **540** and the tray unit **550** can be moved into and out of the apparatus main body **501A** through the one end side of the apparatus main body **501A** while the fixing device **80** is disposed at the one end side of the apparatus main body **501A**. As a result, the user can work to clear a jam, access the fixing device **80**, and operate the transfer unit **540** and the tray unit **550** in one direction. Again, in the present exemplary embodiment, the tray unit **550** passes under the fixing device **80** when being moved from the first inner position to the first outer position. The transfer unit **540** passes under the fixing device **80** when being moved from the second inner position to the second outer position.

(219) The configurations described in the exemplary embodiments can be combined as appropriate.

(220) According to an exemplary embodiment of the present disclosure, the conventional art can be further developed.

(221) While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(222) This application claims the benefit of Japanese Patent Application No. 2022-145891, filed Sep. 14, 2022, and Japanese Patent Application No. 2023-125135, filed Jul. 31, 2023, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image forming apparatus configured to form an image on a recording material, the image forming apparatus comprising: an apparatus main body including a first end provided with a main body opening and a second end opposite to the first end in a horizontal direction; a fixing device configured to heat the recording material; an opening and closing member attached to the apparatus main body, wherein the opening and closing member is configured to be moved from a closed position, where the main body opening is covered, to an open position where the main body

opening is exposed so that the opening and closing member separates from the fixing device; a drawer unit including (i) a first cartridge, (ii) a support member to which the first cartridge is detachably attached, and (iii) a first photosensitive drum disposed on the support member or the first cartridge, wherein the drawer unit is configured to be moved to a first inner position inside the apparatus main body and to a first outer position outside the apparatus main body; and a transfer device configured to be moved to a second inner position inside the apparatus main body and to a second outer position outside the apparatus main body and including a belt, wherein the belt is configured to contact the first photosensitive drum with the transfer device at the second inner position, wherein the fixing device is closer to the first end than to the second end, wherein, when being moved from the first inner position to the first outer position through the main body opening, the drawer unit is configured to be moved in a moving direction intersecting with a rotation axis of the first photosensitive drum, and be moved away from the second end, and wherein, when being moved from the second inner position to the second outer position, the transfer device is configured to be moved away from the second end.

2. The image forming apparatus according to claim 1, wherein, in a state where the opening and closing member is open, the drawer unit is at the first outer position, and the transfer device is at the second outer position, the opening and closing member supports the transfer device, and the transfer device supported by the opening and closing member supports the drawer unit.

3. The image forming apparatus according to claim 1, wherein the drawer unit includes a second cartridge detachably attached to the support member, and a second photosensitive drum disposed on the second cartridge or the support member.

4. The image forming apparatus according to claim 3, wherein, with the drawer unit at the first inner position, the second photosensitive drum is closer to the first end than the first photosensitive drum is, and a rotation axis of the second photosensitive drum is located above the rotation axis of the first photosensitive drum in a vertical direction.

5. The image forming apparatus according to claim 1, wherein, when being moved from the first inner position to the first outer position, the drawer unit is configured to be moved in a direction oblique to the horizontal direction.

6. The image forming apparatus according to claim 5, wherein, when being moved from the first inner position to the first outer position, the drawer unit further is configured to be moved in a direction obliquely upward with respect to the horizontal direction.

7. The image forming apparatus according to claim 1 wherein, at the closed position, the opening and closing member is configured to cover a first path for the recording material traveling toward the fixing device to pass through, and wherein the opening and closing member includes a second path for the recording material that has passed through the fixing device to pass through.

8. The image forming apparatus according to claim 7, wherein, when the opening and closing member is at the closed position, at least a first part of the second path is configured to be located in a first space for the drawer unit to pass through in being moved from the first inner position to the first outer position.

9. The image forming apparatus according to claim 8, wherein, when the opening and closing member is at the closed position, at least a second part of the second path is configured to be located in a second space for the transfer device to pass through in being moved from the second inner position to the second outer position.

10. The image forming apparatus according to claim 1, wherein the transfer device includes a driving roller configured to drive the belt, and a driven roller where the driving roller is located downstream of the driven roller in a direction where the transfer device is moved from the second inner position to the second outer position.

11. The image forming apparatus according to claim 1, further comprising a conveyance unit configured to convey the recording material toward the fixing device, wherein at least a part of the conveyance unit is located in a first space for the drawer unit to pass through in being moved from

the first inner position to the first outer position.

12. The image forming apparatus according to claim 1, wherein the drawer unit is configured to be moved from the first inner position to the first outer position and the transfer device is configured to be moved from the second inner position to the second outer position with the fixing device coupled to the apparatus main body.

13. The image forming apparatus according to claim 1, wherein the drawer unit is configured to pass under the fixing device in being moved from the first inner position to the first outer position, and wherein the transfer device is configured to pass under the fixing device in being moved from the second inner position to the second outer position.

14. The image forming apparatus according to claim 1, further comprising a transfer roller; wherein the belt is configured to contact the first photosensitive drum so that a first transfer portion is formed between the first photosensitive drum and the belt, and wherein the transfer roller is configured to contact the belt so that a second transfer portion is formed between the belt and the transfer roller.

15. The image forming apparatus according to claim 14, wherein, in a state where the opening and closing member is in the closed position, the transfer roller is located inside a space through which the transfer device passes when moved from the second inner position to the second outer position, and wherein, in a state where the opening and closing member is in the open position, the transfer roller is located outside the space.

16. The image forming apparatus according to claim 1, wherein, when the drawer unit is moved from the first inner position to the first outer position, a moving direction of the drawer unit changes.

17. The image forming apparatus according to claim 1, wherein, when the transfer device is moved from the second inner position to the second outer position, a moving direction of the transfer device changes.
